Rockchip RKStudio_Tool User Guide

Document Identifier: RK-XX-XX-nnn

Release Version: V1.1.2

Date: 2025-02-06

Classification: □ Top Secret □ Confidential □ Internal ■ Public

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Preface

Overview

This document is a use guide for RKStudio_Tool, covering features such as graphical development and real-time parameter tuning.

Product Version

Chip Name	Core Version
RK2118/RK2116/RK2108/RK3308/RK3506/RK356X	This document is unrelated to the kernel and serves solely as a reference document for the `RKStudio_Tool algorithm control`

Intended Audience

This guide is primarily intended for the following engineers:

- Audio Engineers
- Technical Support Engineers

Revision History

Version	Author	Date	Change Description
V1.0.0	Huaping Liao	2024-02-22	Initial version
V1.1.0	Huaping Liao	2024-05-16	Add wav partition and sub-map description
V1.1.1	Di Jiang	2024-12-17	Parameter import/export instructions
V1.1.2	Di Jiang	2025-02-06	Updated diagrams and operation instructions
V1.1.3	Huaping Liao	2025-08-19	Updated diagrams and operation instructions
V1.1.4	Huaping Liao	2026-02-02	Updated diagrams and operation instructions

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1 Hardware and board-side Firmware Preparation

1.1 Hardware Preparation

Currently supports development boards for RK2118 and RK3308 chips.

1.1.1 RK2118

Taking the RK_AUTOMOTIVE_DSP_DEMO_RK2118M_V10 development board as an example, the relevant interfaces are shown in the figure below:

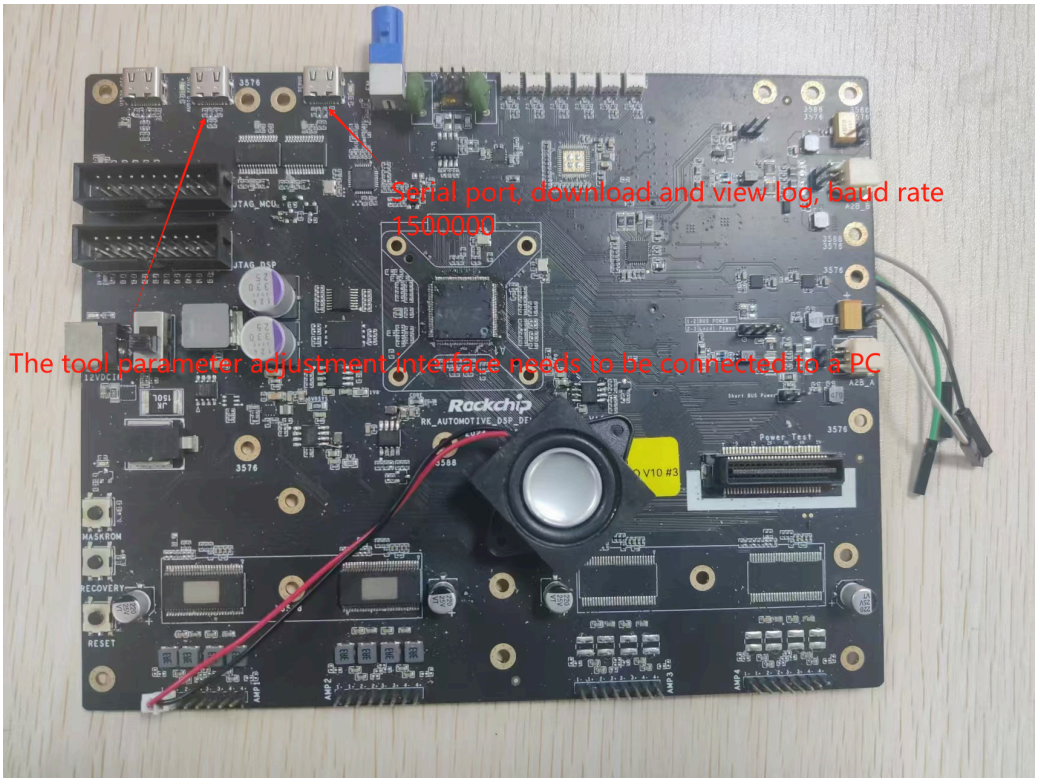


Figure 1.1-1 RK2118M Development Board

1. Debugging serial port on the board (marked as DEBUG on the board), for viewing debugging information, baud rate: 1500000

2. Tool tuning interface (marked as AUDIO DEBUG on the board), which needs to be connected to the PC, also using serial communication protocol, with a baud rate of 1500000

When in use, connect the tool tuning interface (AUDIO DEBUG) to the computer using a Type-C cable and ensure the development board is powered on. The other debugging serial port (DEBUG port) is mainly for viewing debugging information and does not affect the actual tuning if not connected.

1.1.2 RK3308

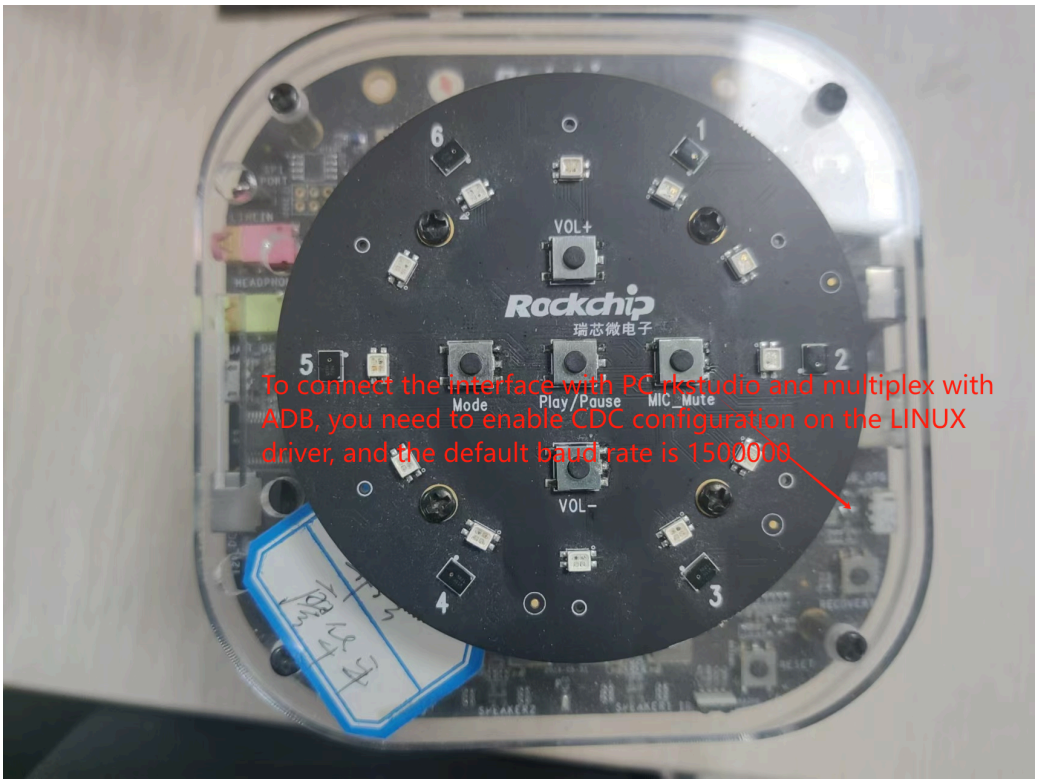


Figure 1.1-2 RK3308 Development Board

1. Interface for connecting RKStudio on RK3308, shared with ADB, requires a LINUX driver to enable CDC configuration, default baud rate is 1500000
For RK3308 development boards, connect the USB-OTG port on the board to the computer using a micro-USB cable, and ensure that the RK3308's CDC serial port device is effective. Confirm that there is a /dev/ttyGS0 device node on the board.

1.2 Board-side Firmware Preparation

Since many firmware versions do not enable the tool tuning interface by default, configuration is required.

- For RK2118 development boards, go to the RK2118 RTT development environment, ensure that RT_USING_UART2 is enabled in menuconfig, and set CONFIG_RT_SERIAL_RB_BUFSZ to 1024. Then recompile and flash the firmware.
- For RK3308 boards, configure the environment variable rkstudio_tuning=on to enable debugging mode.

2 Tools and Connection

2.1 Tool Download

The tool is generally released with the SDK, located in the `<sdkPath>/bsp/rockchip/tools/` directory. The tool package name starts with **RKStudio**. The tool does not require installation; simply extract and run it directly. The system environment must be **Windows 10 or later.**

2.2 Connection with the Board

Open the tool, connect the tool tuning interface of the board to the computer, and ensure the development board is powered on. Note that the **AUDIO DEBUG** port should be connected, not the DEBUG port on the board. The current connection between the tool and the board is through a serial port, which can be viewed in the Device Manager. The tool does not automatically connect by default. Click on the menu bar in the middle of the tool, select Device -> choose the corresponding COM port, set the baud rate to 1500000, and then click Connect. At this point, the status bar on the right will display "Serial port opened, received board information: Chip signal: rk2118," etc., indicating a successful connection, and you can proceed with audio stream deployment and tuning.

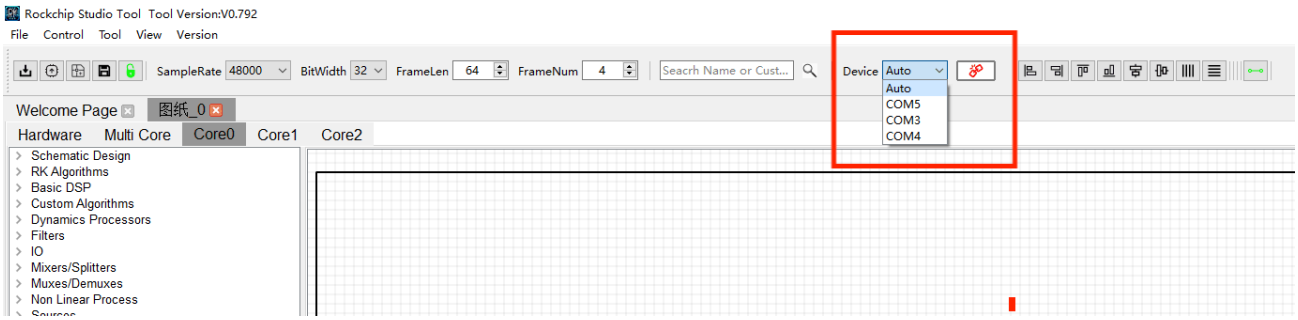


Figure 2.1-1 Serial Port Connection

3 Audio Path Deployment

3.1 Audio Path Deployment

In the project page, select a core that needs to be deployed, usually core 0, and drag out the components as shown in the figure below:

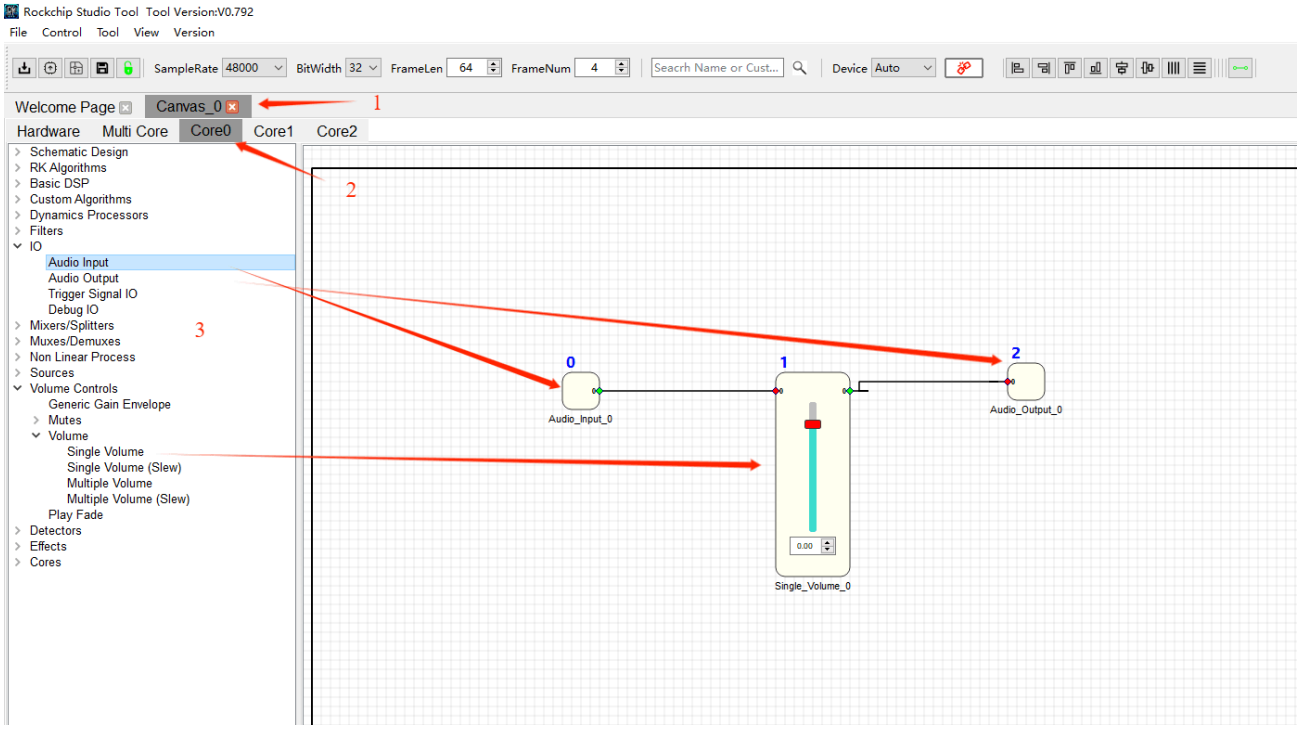


Figure 3.1-1 Component Deployment

1. You can use the default diagram or create a new one.
2. Choose the core that needs to be deployed; all three cores can be deployed.
3. Drag the components from the list into the deployment diagram.
4. Wiring:

The IO in the component bar, i.e., Input and Output, refers to the audio input and output interfaces of the algorithm process, which are independent of the actual hardware. If only the internal source of the algorithm is needed, you can use the Source-related components, and there is no need to use the Input component. Connect the components with lines to complete the audio component connection.

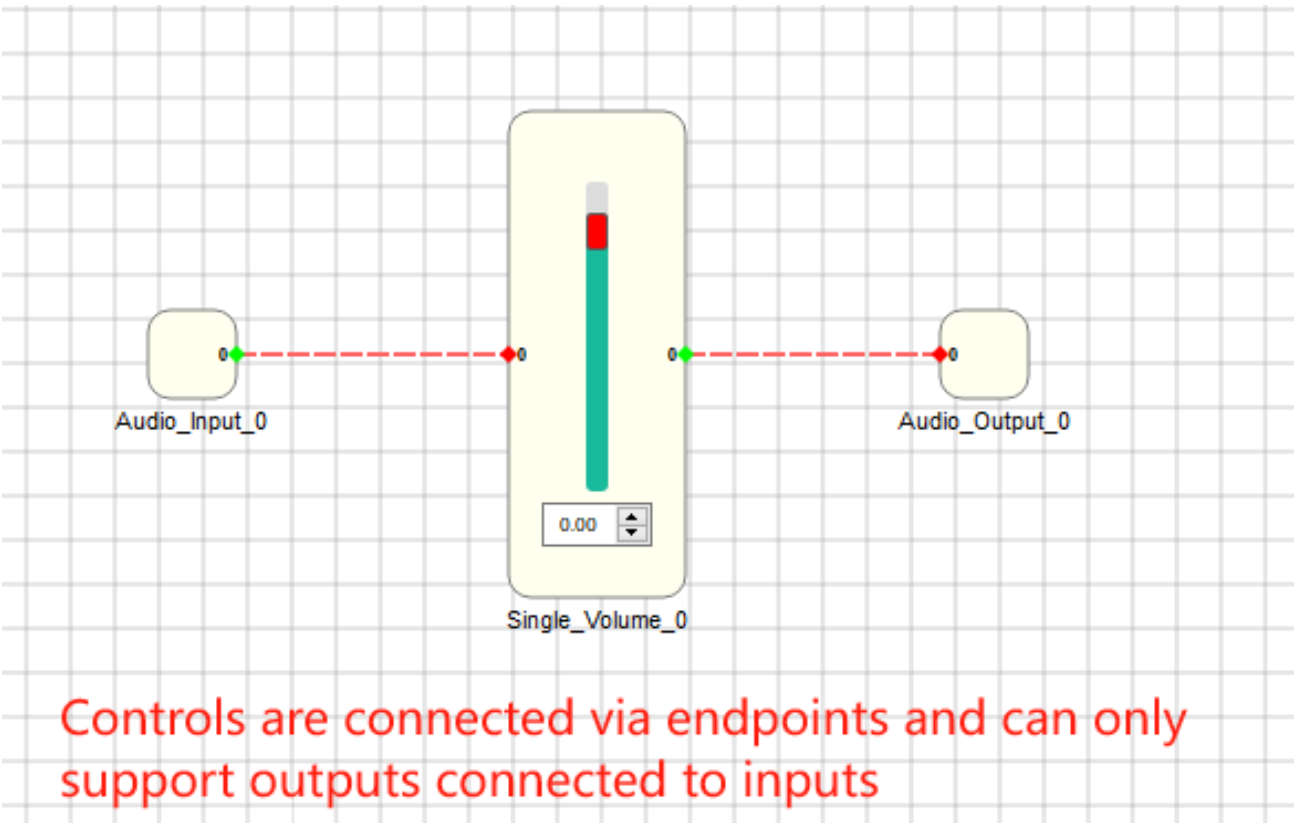


Figure 3.1-2 Component Connection

Components are connected through endpoints and only support connecting outputs to inputs.

5. Adding or Removing Channels and Other Operations

   If you need to add or reduce the number of channels, you can right-click on the component, and a menu will pop up, allowing you to add or reduce the number of channels, delete, etc.

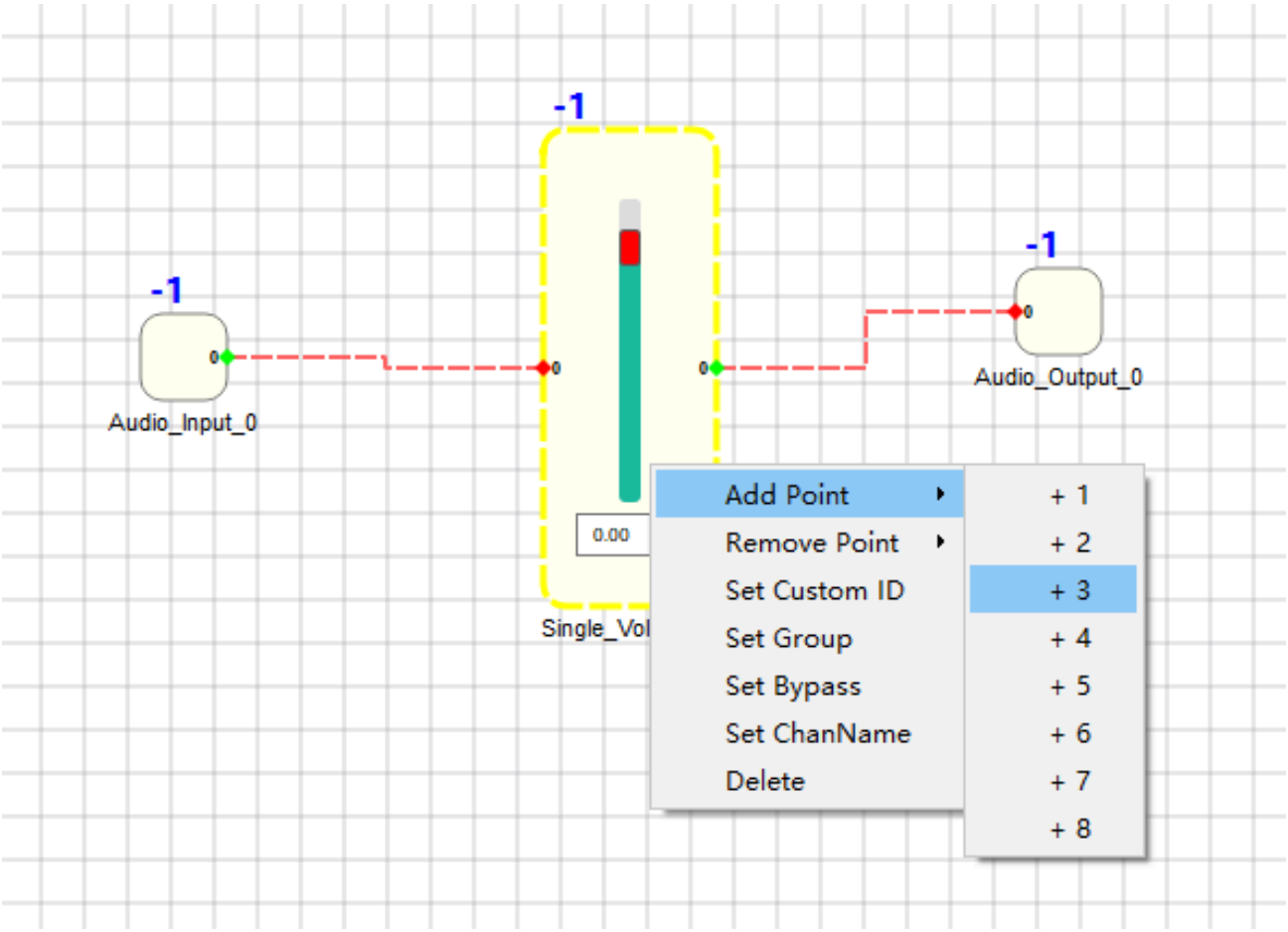


Figure 3.1-3 Right-Click Menu

It also supports copy (Ctrl+C), paste (Ctrl+V), and delete (Delete) operations.

3.2 Downloading and Debugging

After the path deployment is complete, click the download shortcut key to download directly.

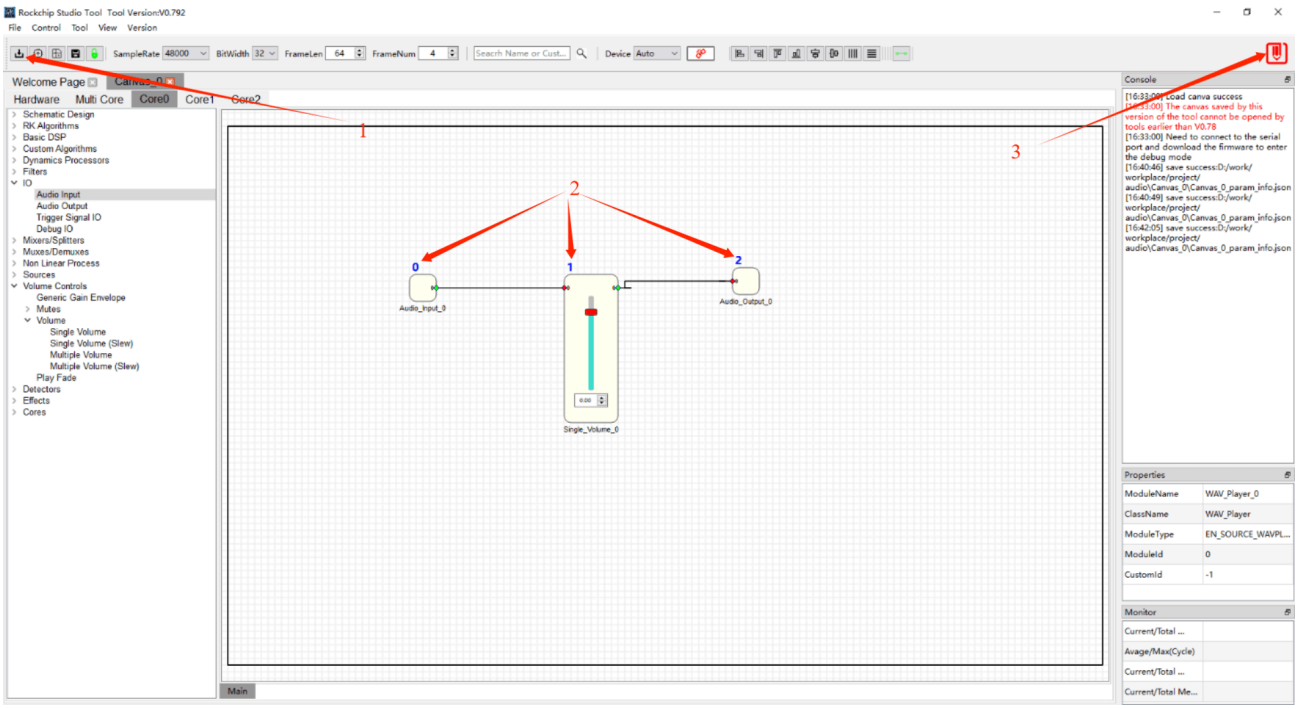


Figure 3.2-1 Right-Click Menu

1. Download shortcut key
2. The component will display a unique idx , which is also the execution order of the component
3. The current status is displayed as designing (pencil shape); after clicking download, it will automatically change to debugging status

When clicking Save Project or Download, if there are unconnected endpoints (except for Input which can be left unconnected), an error will pop up, and the focus will be on the erroneous module:

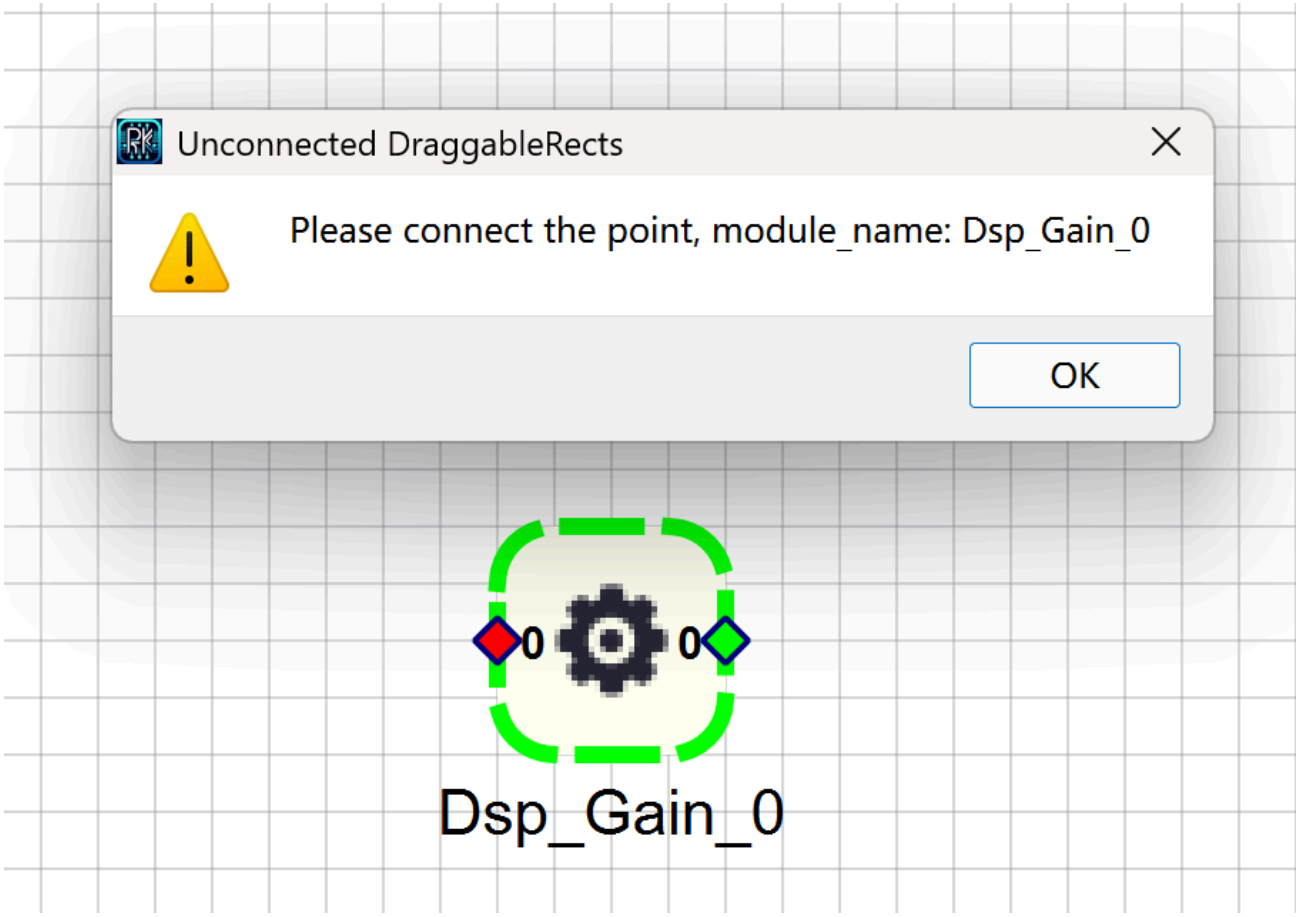


Figure 3.2-2 Connection Integrity Error

At this point, you can directly broadcast the input audio.

You can adjust the audio through the Single_Volume component by simply dragging the audio control bar. Some component parameters cannot be set directly through the chart, such as the Gain module. You can double-click the left mouse button, which will pop up a settings interface. Enter the parameters, press Enter, switch to another parameter field, or click OK, and the settings will take effect immediately.

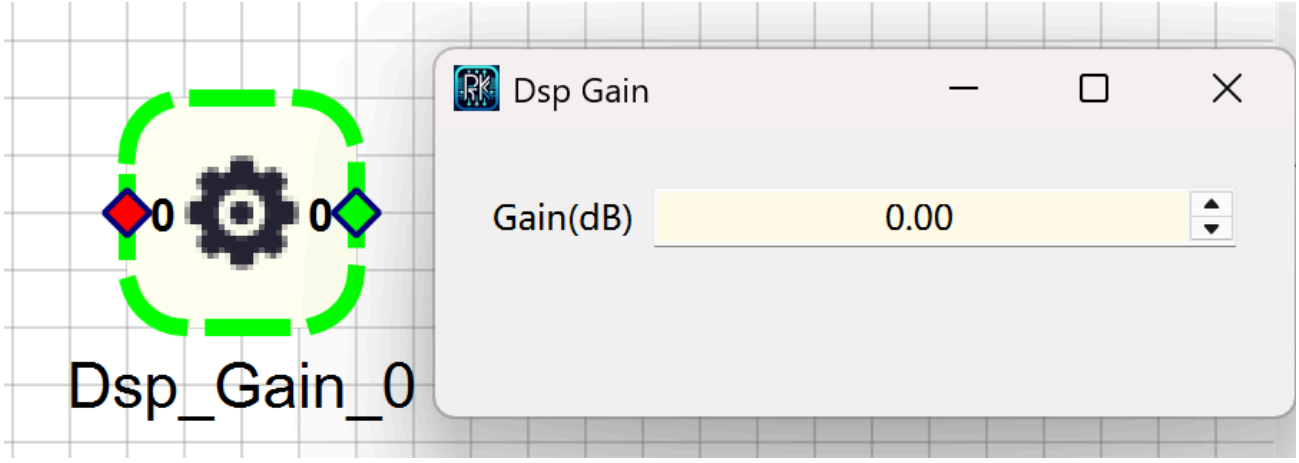


Figure 3.2-3 Double-Click to Open Settings Interface

4 Basic Tool Function Introduction

The tool interface can be divided into the following areas:

The following numbered sequence is from top to bottom and from left to right.

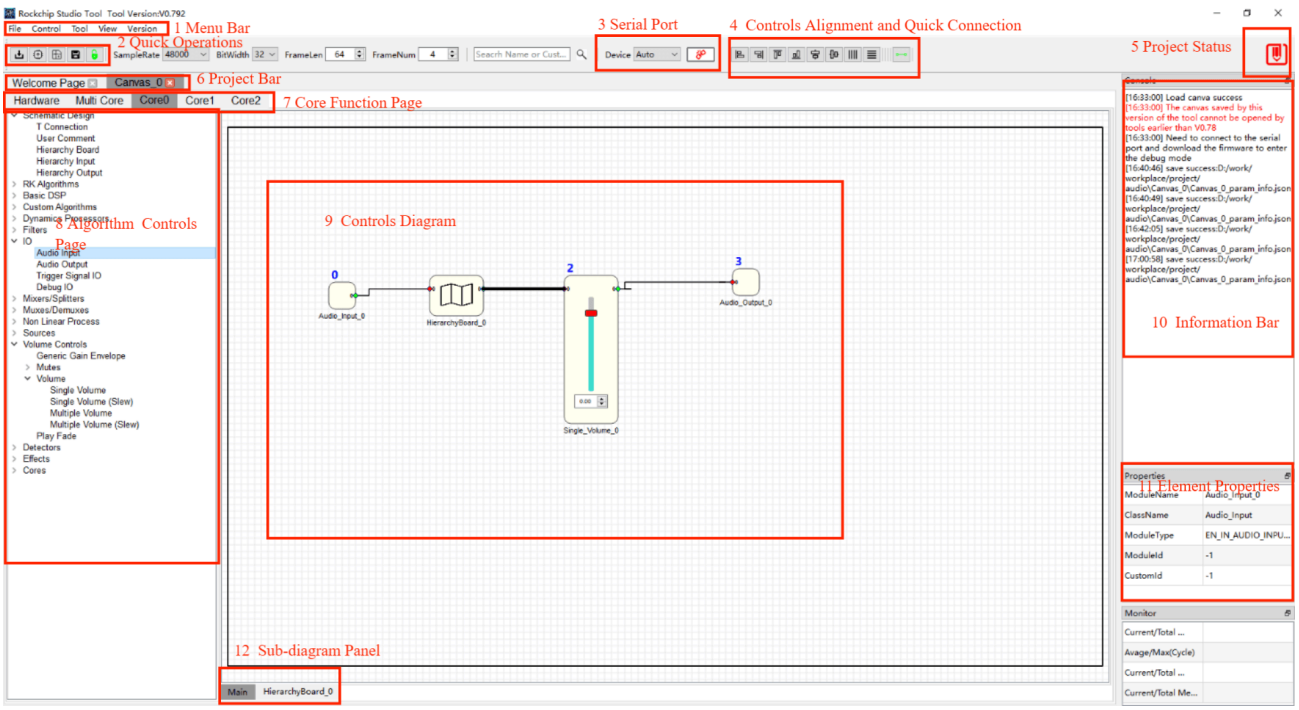


Figure 4-1 Tool Interface Introduction

1. Menu Bar: Contains File, Control, Tool, View, Debug, Version
2. Quick Operations: Includes Download, Save Firmware, Save Diagram, Save Project, Lock/Unlock Diagram
3. Serial Port Connection Settings
4. Controls Alignment and Quick Connection
5. Project Status: Displays the current working status, with a pencil shape indicating Designing
6. Project Bar: Click to quickly switch between different project interfaces
7. Core Function Page: Includes Hardware Configuration, Multi-core Connection, Core 0, Core 1, and Core 2 deployment pages.
8. Algorithm Controls Page: Lists the currently supported algorithm Controls, which can be dragged out by the user.
9. Controls Diagram: The main audio stream deployment page.
10. Information Bar: Displays current debug information and interaction information with the board.
11. Element Properties: Displays basic information about the selected element
12. Sub-diagram Panel: Displays the current main diagram and sub-diagrams.

Note: The above diagram interface version number is V0.792, and the layout may be slightly adjusted in the new version after updates!

4.1 Menu Bar

The tool's menu bar includes File, Control, Tools, View, Debug and Version.

4.1.1 File

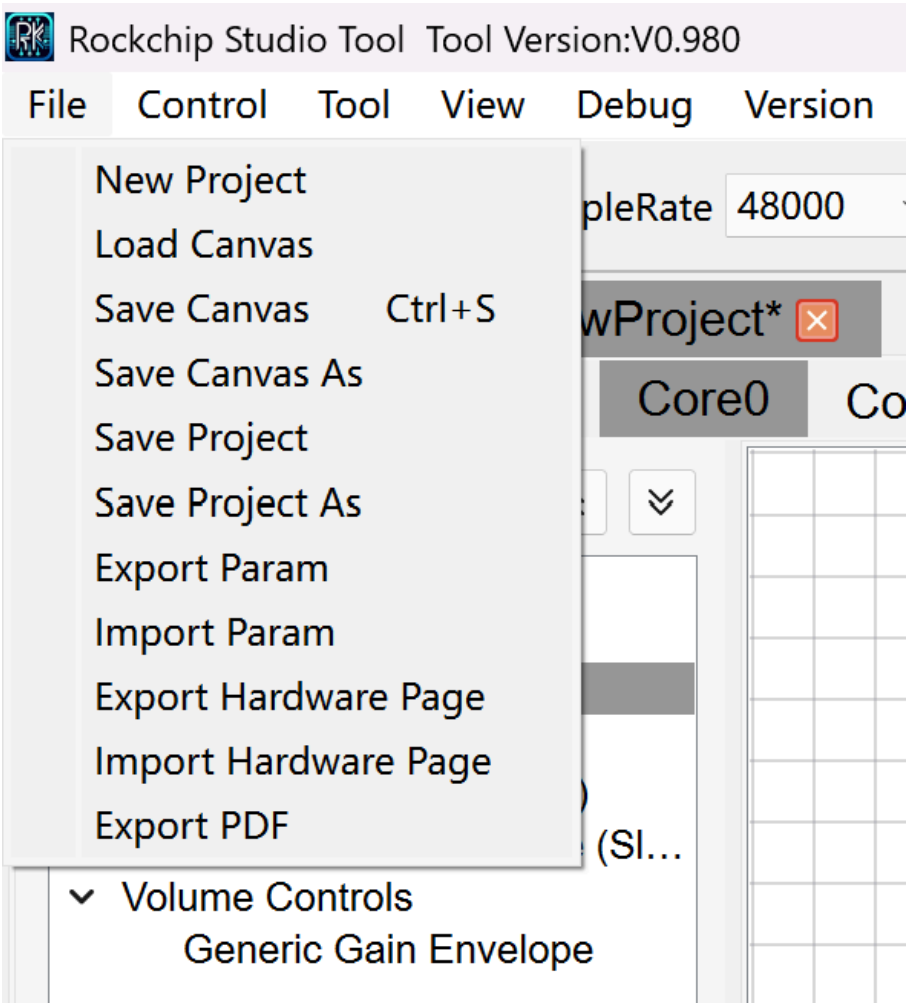


Figure 4.1.1-1 File Function Introduction

The File function includes the following:

- New project: Create a new blank project, with the default project name in the form of <Diagram_Number>.
- Load Canvas: Load a previously saved project file, which will be displayed as the project name in the project bar.
- Save Canvas: Save the current project, which can be an incomplete or complete project. Saving will generate a .json file, and the saved file name will be the project name.
- Save Canvas As: Save the current diagram to a new path.
- Save Project: Save the current complete project. Unlike saving a diagram, this requires the project to have complete connectivity to succeed. The saved files include the .json project file, the .h

parameter header file, and the .bin firmware file.

- Save Project As: Save the current project to a new path.
- Export Param: Export the parameters of the diagram as a .json file.
- Import Param: Import the parameter configuration of a .json file into the current diagram.
- Export Hardware: Exports hardware components from the Hardware Page as a .json file for independent hardware management.
- Import Hardware: Imports a .json hardware configuration file into the drawing.
- Export to PDF: Exports the current design as a PDF document for presentation, review, or printing.

4.1.1.1 Parameter Import and Export

In addition to saving and loading complete design diagrams, the tool also supports exporting module parameters as a .json file separately, which is convenient for users to handle scenarios where the same diagram has different parameter configurations.

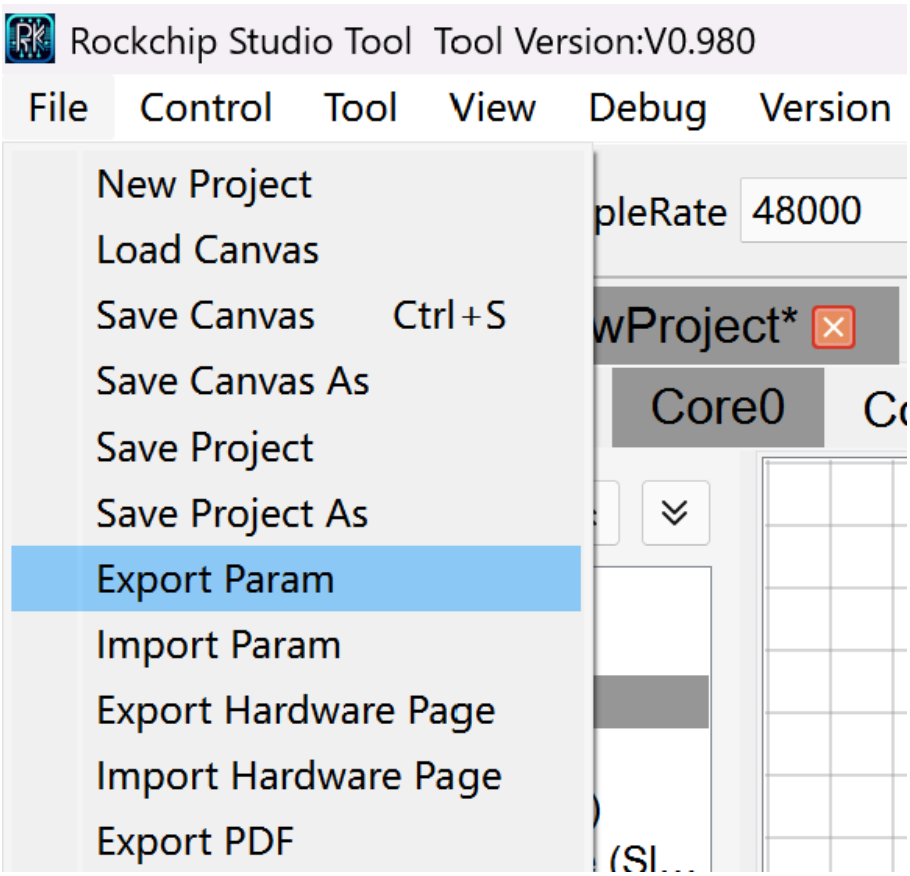


Figure 4.1.1.1-1 Export Current Diagram Parameters

The .json file is not only convenient for users to compare parameters but can also be loaded into the current diagram through the "File - Import Parameters" function.

Note that parameter import is matched based on module names, and if the module name changes, it may prevent the desired module parameters from being imported successfully.

4.1.2 Control

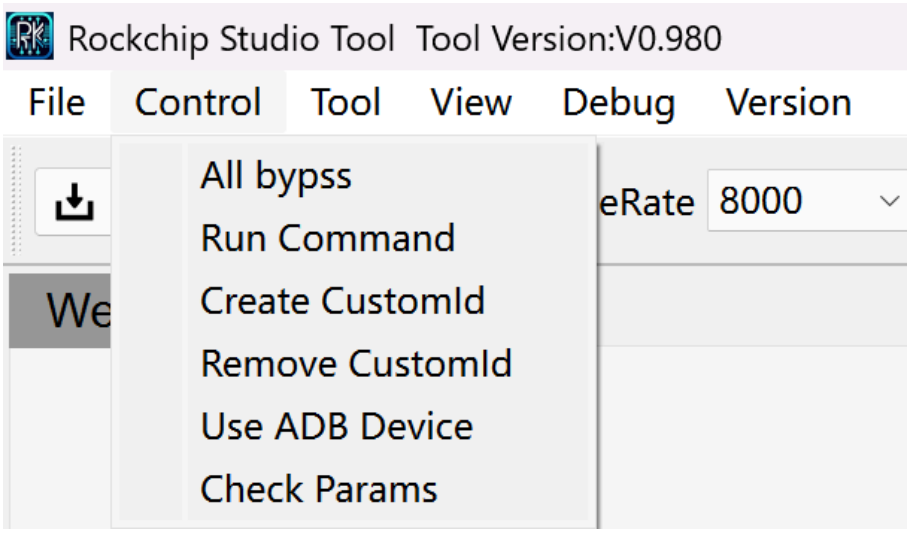


Figure 4.1.2-1 Control Function Introduction

The Control function includes the following:

- All bypass: Directly bypasses the entire RKStudio algorithm process, allowing input to reach output directly.
- Run Command: A command input box will pop up, allowing you to enter commands and execute them on the board, such as ls/help, etc.
- Create CustomId: Creates custom IDs for all components with one click to distinguish between them. If a component already has an existing custom ID, the original ID will be retained.
- Remove CustomId: Removes custom IDs from all components with a single click.
- Use ADB Device: Establishes a communication connection with the target board via the ADB (Android Debug Bridge) interface.
  Check Params: Performs an automated global logic validation on the current design. This function checks for common logic errors—such as floating pins, component parameter conflicts, or duplicate Custom IDs—to ensure the drawing is valid and functional before execution or firmware generation.

4.1.3 Tool

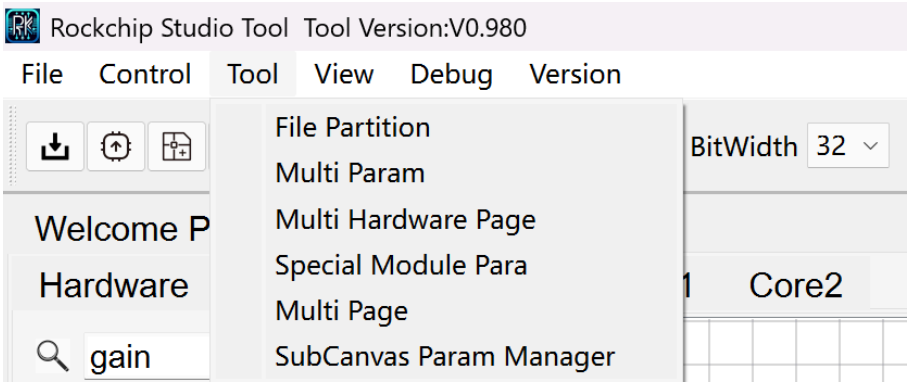


Figure 4.1.3-1 Tools Function Introduction

The Tool function includes the following:

- File Partition: Allows you to allocate a storage space for storing audio files and set an index, which can be used in conjunction with wav_player components.
- Multi Param : Allows you to divide components into several different groups, each of which can independently import parameters from other diagrams.
- Multi Hardware Page: Enables the management of multiple hardware pages.
- Special Module Para: Allows for parameter management of individual modules.
- Multi Page: Enables the management of multiple drawing pages, where each page can have a different signal path.
- SubCanvas Param Manager: Allows for parameter management of sub-canvas modules.

4.1.3.1 File Partition

Example of using file partition with wavPlayer component

If the wavPlayer component is used, click Tools -> WAV File Configuration and select the corresponding WAV file. The folder containing the main files should be placed in the WavData directory of the tool.

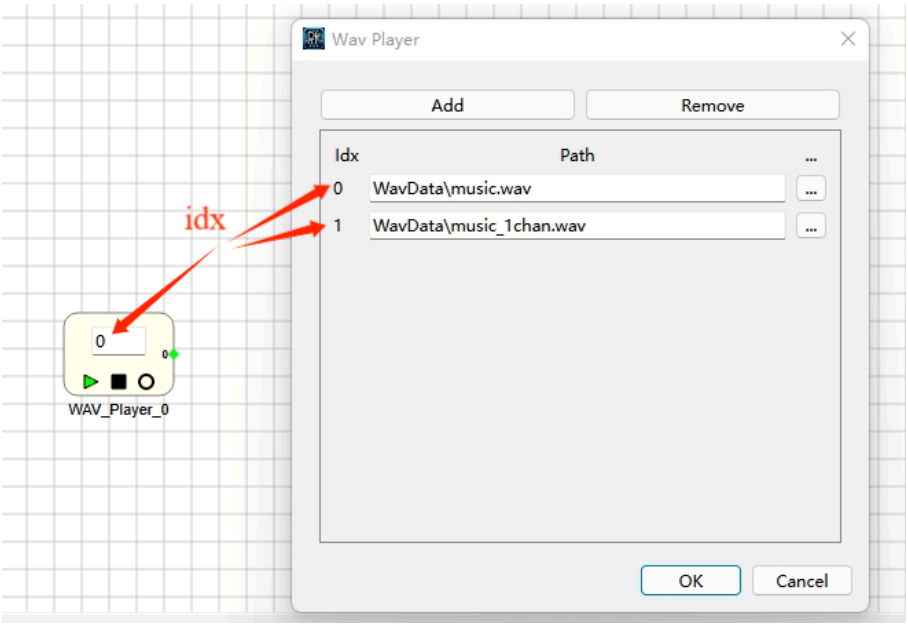


Figure 4.1.3.1-2 WAV File Configuration

After selecting the file and clicking OK, the tool will automatically package the selected WAV file into an IMG image file. A pop-up window will inform you of the generated IMG file path, which needs to be burned to the rkwav partition on the flash. The idx in the configuration page corresponds to the idx of the wavplayer player, which will play the corresponding WAV file based on the idx. Currently, WAV and PCM audio file formats are supported and can be configured in the wav_player configuration page.

4.1.3.2 Multi Param

Group corresponds to the classification of plugins, and None defaults to all plugins in the entire diagram. If you want to group control of plugins, for example, dividing Source_Chime and Source_Sine into part2, right-click on the component, select Set Group, and choose part2 to assign the component to the part2 group.

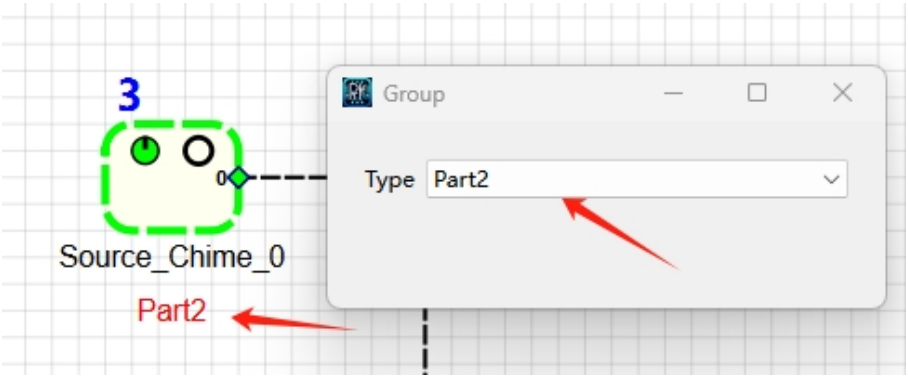


Figure 4.1.3.2-1 Configure Component Group

Then, through "Tools - Multi-Group Parameter Configuration," you can quickly import the saved parameter files. Click "...", select the exported parameter .json file, and click the Import button to

quickly import the corresponding parameter configuration, which will take effect on the diagram and be configured to the board.

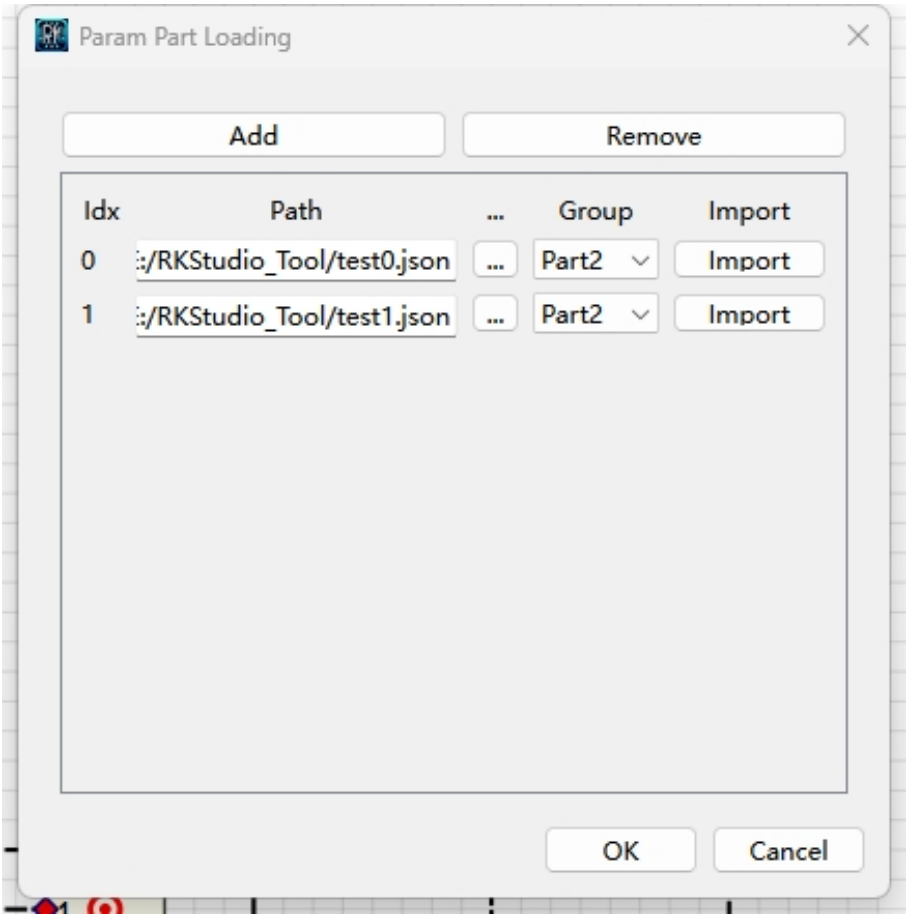


Figure 4.1.3.2-2 Configure Multi-Group Parameters and Quickly Import

During the firmware packaging process, the configuration parameters of the plugins corresponding to the configured group will be packaged into the firmware (real-time download will not package). Only the plugins corresponding to the current group will be packaged. Then, on the board, you can switch to the corresponding group with idx 0 by using the command: rkstudio_group_id_set 0, and it will take effect.

4.1.3.3 Multi Hardware Page

Export the hardware page JSON via **File - Export Hardware**, then enter the management interface via **Tool - Multi Hardware Page**. Click the **...** button, select the hardware JSON file you wish to manage, and click **Import**. This will quickly import the corresponding hardware page and apply the configuration to the drawing in real-time.

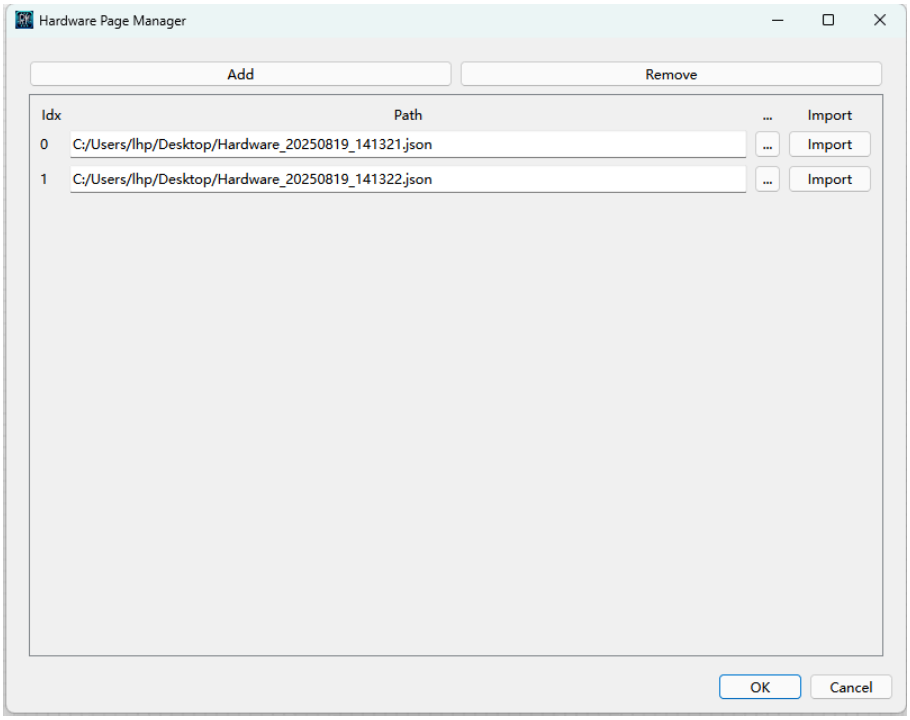


Figure 4.1.3.3-1 Configure multiple hardware setups and import quickly

During the firmware packing process, the configured hardware information will be bundled into the firmware (note: real-time downloading will not trigger packing). On the board side, you can switch to the hardware drawing with idx 0 and make it effective by using the command:
`rkstudio_group_hw_set 0` .

4.1.3.4 Special Module Para

Select an individual parameter module, right-click and choose **Export Param** to export the module configuration as a JSON file. Then, enter the management interface via **Tool - Special Module Para**. Click the ... button, select the parameter JSON, and click **Import**. This will quickly import the corresponding parameter configuration and apply it to the drawing in real-time.

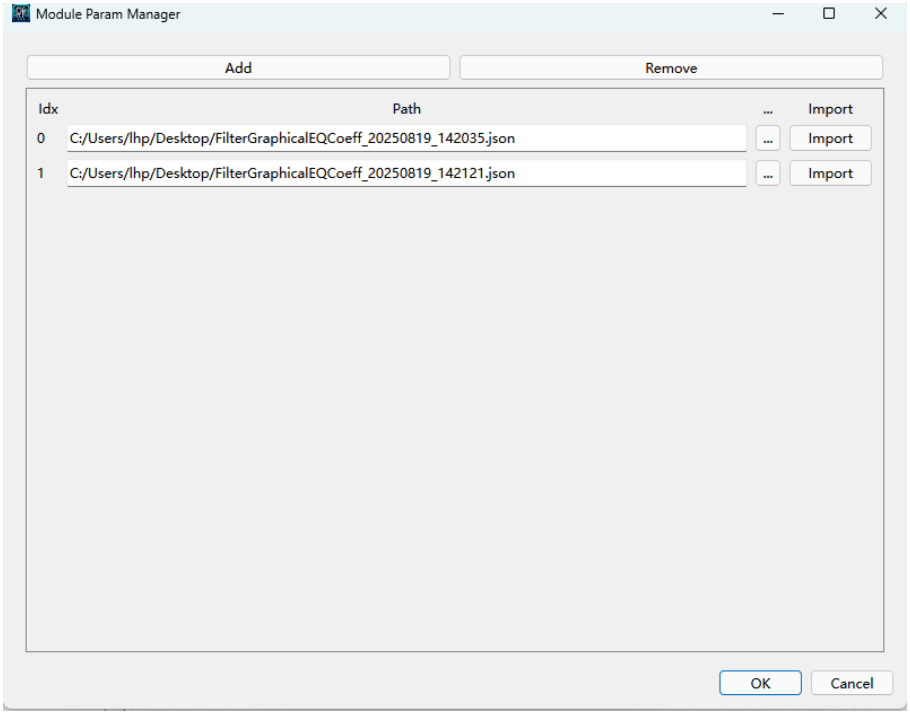


Figure 4.1.3.4-1 Configure individual parameter paths and import quickly

During the firmware packing process, the configured module parameter information will be bundled into the firmware (note: real-time downloading will not trigger packing). On the board side, you can switch to the module parameters with idx 0 and make them effective by using the command: `rkstudio_custom_param_set 0` .

4.1.3.5 Multi Page

The **.rkproj** file is the project drawing file. To manage and switch between multiple drawings simultaneously, enter the management interface via **Tool - Multi Page**. Click the ... button, select the drawings to be managed, and click **Import**. You will then see a new drawing page opened in the Project Bar, loading the corresponding project file.

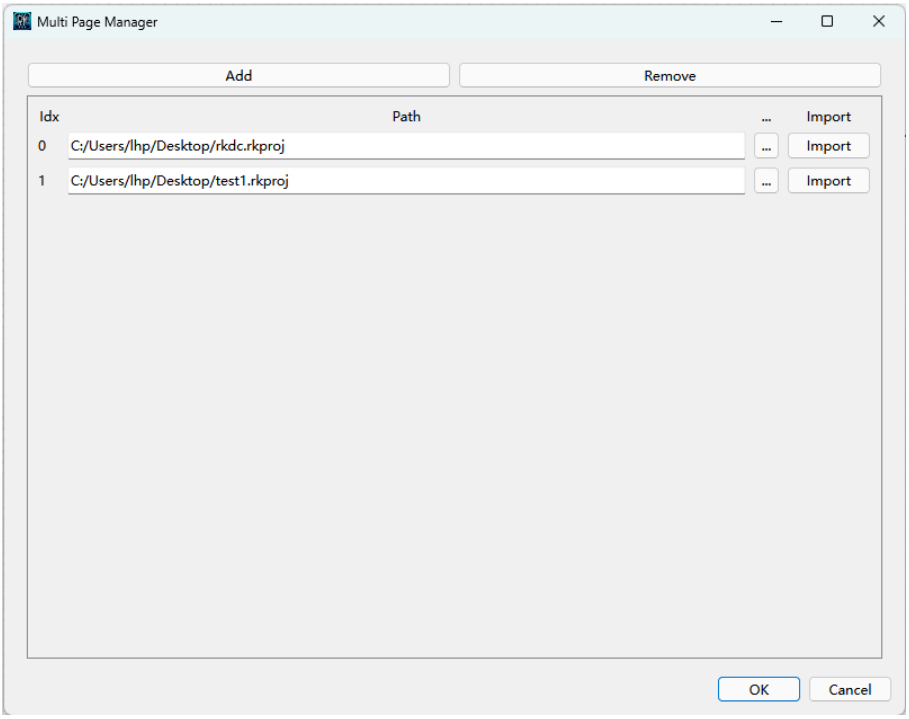


Figure 4.1.3.5-1 Configure multiple drawing paths and import quickly

During the firmware packing process, information for multiple drawings will be bundled into the firmware (note: real-time downloading will not trigger packing). On the board side, you can switch to the drawing with idx 0 by using the command: `rkstudio_group_fw_set 0` . A fade-in/fade-out effect will be applied during the transition to prevent POP noise.

4.1.3.6 SubCanvas Param Manager

Sub-canvas parameter files (typically in .json format) are configuration files for algorithm modules. To manage and switch between multiple sets of sub-canvas parameters, enter the **Module Param Manager** interface via **Tool - SubCanvas Param Manager**. Click **...** to select the corresponding parameter file, then click **Import** to load the parameters into the corresponding Hierarchy level of the project.

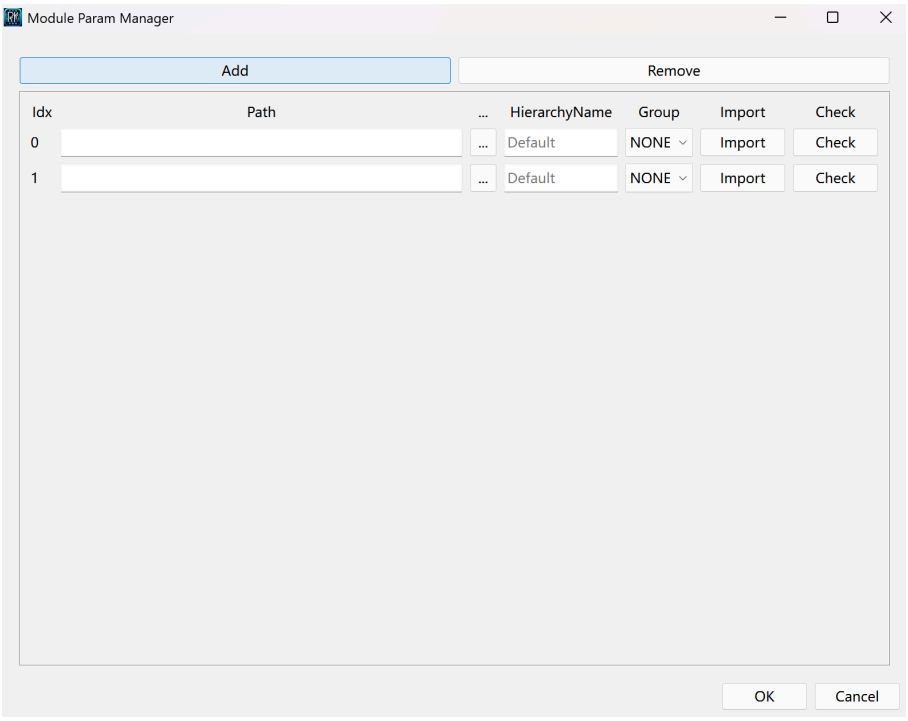


Figure 4.1.3.6-1 Sub-canvas parameter management and quick import

During the firmware packing process, multiple sets of sub-canvas parameter information will be unified and bundled into the firmware. While running on the board, use the command: `rkstudio_group_fw_set [idx]` (where idx corresponds to the index number on the left side of the UI) to switch to the corresponding parameter set in real-time. The system supports fade-in/fade-out processing during switching to ensure smooth audio transitions and prevent POP noise.

4.1.4 View

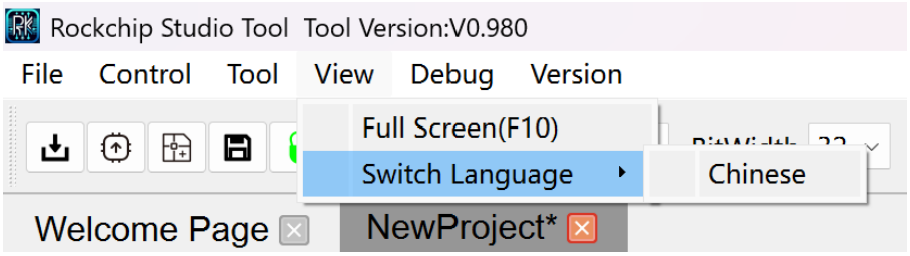


Figure 4.1.4-1 View Function Introduction

The View function includes the following:

- Full Screen（F10）: Clicking or pressing F10 will enter full-screen mode for the deployment diagram.
- Switch Language : Switch between Chinese and English.

4.1.5 Debug

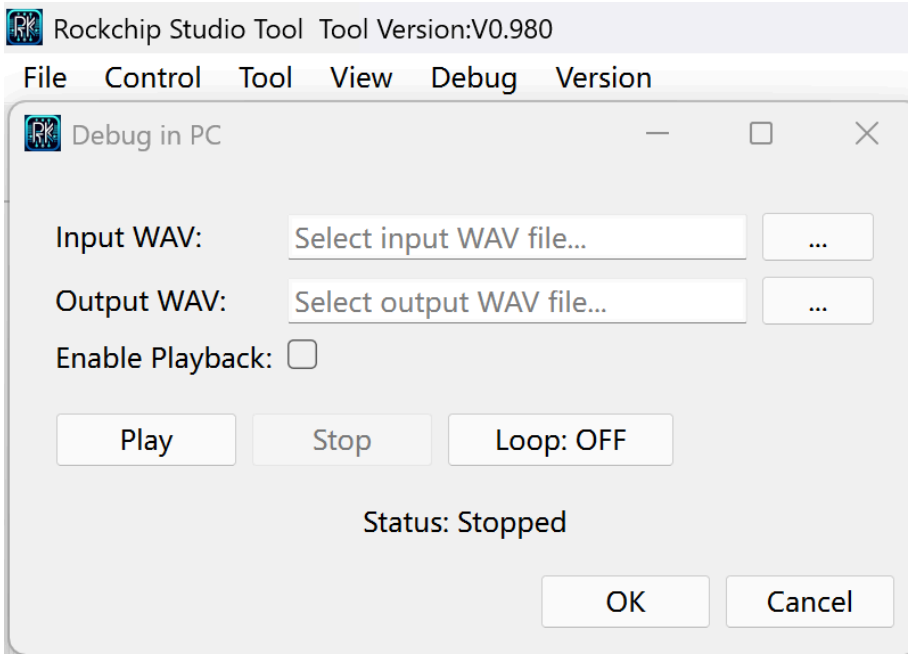


图 4.1.5-1 Debug Function Introduction

The Debug function includes the following:

- Debug in PC
  This feature allows users to simulate the audio processing flow within a PC environment:
  - Input/Output WAV Selection: Load a local .wav file as the simulation source and specify an output path to save the processed audio for data comparison and analysis.
  - Enable Playback: When selected, the simulated audio results are routed directly to the PC's sound card and speakers, allowing users to monitor the algorithm's effects in real-time without external hardware.
  - Control & Monitoring: Supports Play, Stop, and Loop mode toggling, while displaying the real-time execution status of the simulation task.

4.1.6 Version

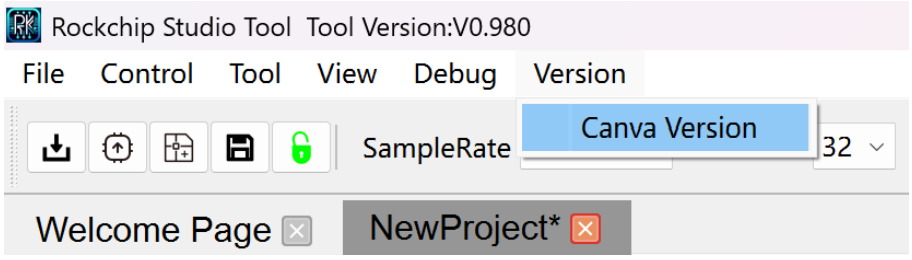


Figure 4.1.6-1 Version Function Introduction

The Version function includes the following:

- Canvas Version: Set the diagram version number.

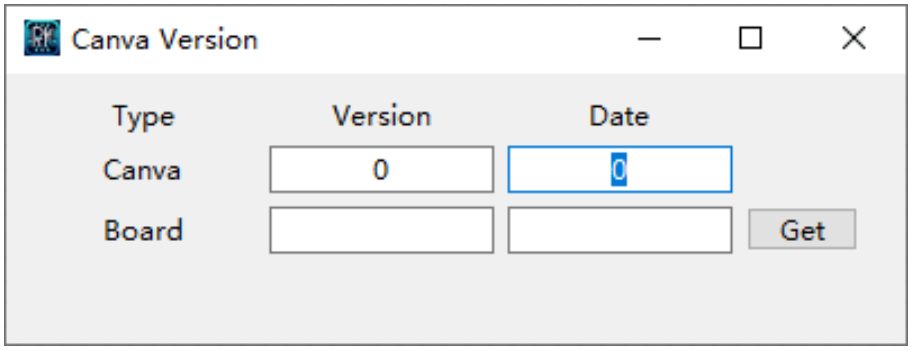


Figure 4.1.5-2 Set Diagram Version Number

4.2 Quick Operation Bar and Project Status Information Bar

The icons in the Quick Operation Bar from left to right are as follows:

- Firmware Download: Download the current firmware to the board. When initially laying out or modifying the layout, you need to download the firmware to the board to proceed with audio stream deployment and tuning. This is effective in real-time and will be lost upon power-off. If the audio stream is deployed and you want it to take effect immediately after power-on, you can use the Save Firmware function to obtain the .bin firmware file and deploy it to the board's firmware.
- Save Firmware: For convenience, save the firmware .bin file to the project directory, with the file name rkstudio.bin.
- Save Diagram: Unlike firmware saving, this does not check connection validity and does not generate a .bin file. It only saves the diagram's .json file.
- Save Project: Save the current complete project. Unlike saving a diagram, this requires the project to have complete connectivity to succeed. The saved files include the .json project file, the .h parameter header file, and the .bin firmware file.
- Lock/Unlock Diagram: Click to toggle the state. The diagram cannot be edited in the locked state.

4.3 Serial Port Connection Settings

Select the corresponding COM port of the serial port on the right side, set the baud rate, and then click the button on the right to connect/disconnect;

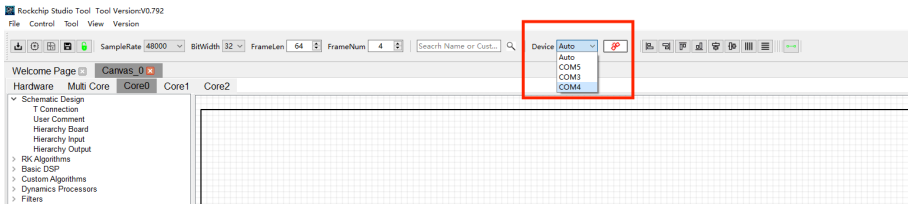


Figure 4.3-1 Serial Port Connection

4.4 Controls Alignment and Quick Connection

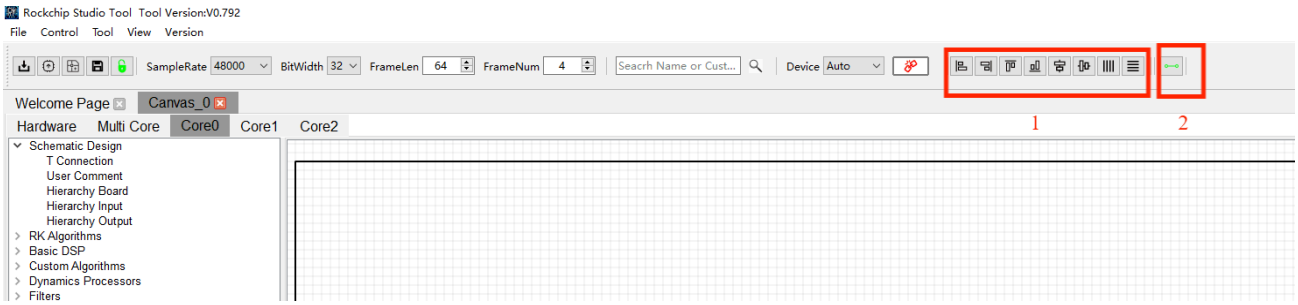


Figure 4.4-1 Component Alignment and Quick Connection

1. Set component alignment or expansion. Select components with the left mouse button and then click the corresponding alignment or expansion method.
2. Quick connection: Select the corresponding two components and click Quick Connection to connect the input and output of the two components.

4.5 Project Status

The status information bar displays the current working status, including:

- Designing: The diagram design phase, represented by a pencil icon. This is the initial deployment or when modifications are made to the diagram, such as adding or deleting components, adding parameter connection lines, or adding or deleting parameter counts.
- Debugging: The debugging stage, which is entered after the diagram deployment is complete and the firmware download is successful. During this stage, you can adjust component parameters for real-time tuning, which will take effect immediately. If modifications are made to the diagram, it will automatically return to the Designing stage.

4.6 Project Bar

You can open multiple project diagrams simultaneously and quickly switch between them by clicking on different project pages.

4.7 Core Function Page

The Core Function Page includes hardware configuration, multi-core connection, and deployment pages for Core 0, Core 1, and Core 2.

4.7.1 Hardware Configuration Page

Configure and deploy hardware connections. In the hardware configuration page, the list on the left will display the currently supported hardware configuration interfaces. Drag out the hardware interfaces and double-click to configure them. When input and output components are added to the core page, they will also be displayed in the hardware configuration page, where they can be connected to hardware interfaces.

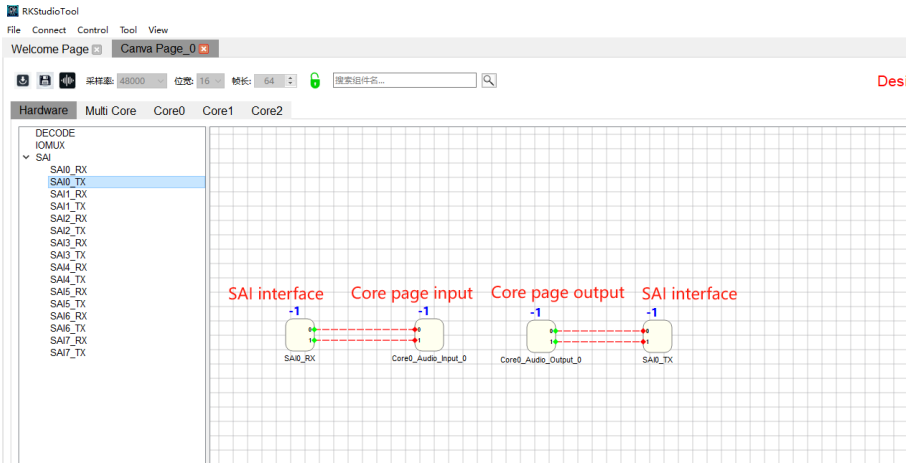


Figure 4.7.1-1 Hardware Configuration

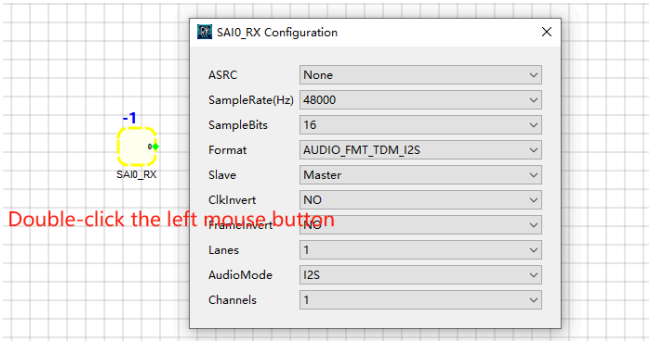


Figure 4.7.1-2 SAI Configuration Page

Double-click to display configuration

4.7.2 Multi-Core Connection

Deploy the connection relationships of multiple cores. In the multi-core connection page, if Core components are deployed, they will be displayed on the page. You can deploy multi-core data streams on this page. Double-clicking on the Cores component on the core page allows you to configure the data synchronization method. Setting it to No means no synchronization, while setting it to Sync means synchronization. After configuring to Sync, it will wait for data from another core to arrive before proceeding with subsequent processing; otherwise, it will wait.

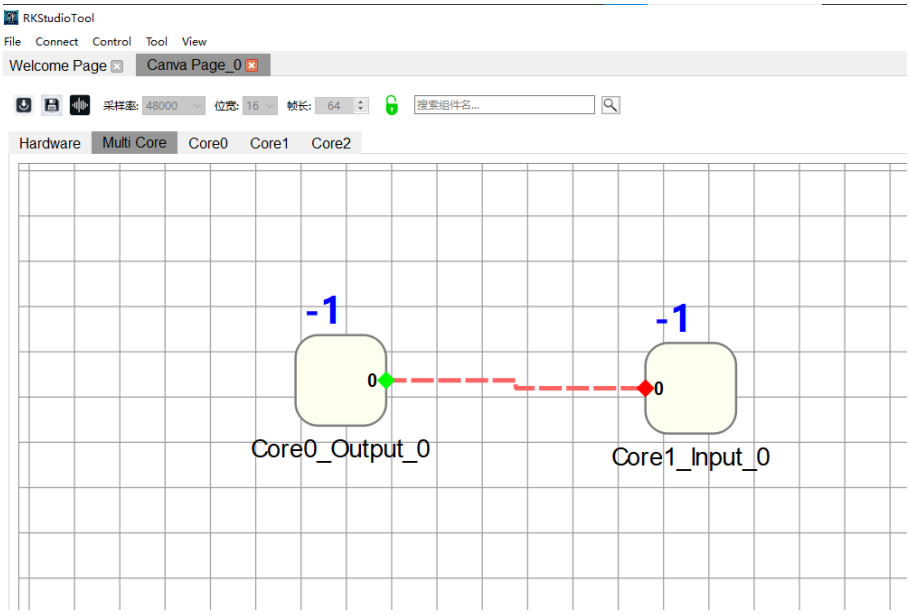


Figure 4.7.2-1 Multi-Core Connection Deployment

4.7.3 Core Configuration Page

The core configuration page will display the current core pages based on the chip configuration. For example, if RK2118 has three DSP cores, then three core pages will be displayed here.
The core page includes the algorithm component page and the component diagram, and you can configure by dragging the components from the left to the component diagram.

4.8 Algorithm Controls Page

Deploy the basic audio algorithm data flow, which is also the most important functional page. The left side lists the supported algorithm components, with IO for algorithm input and output components, and Cores for multi-core input and output components. It also supports basic audio effects algorithms such as EQ, IIR, FIR, Mixers/Splitters, Muxers/Demuxers, Delay, Compressor, Limiter, DRC, etc. For specific algorithm components and parameter introductions, refer to the document <Rockchip_RKStudio_Tool_Algorithm_Guide_CN.pdf>. The right side is the current component wiring diagram, where you can drag components from left to right for layout.

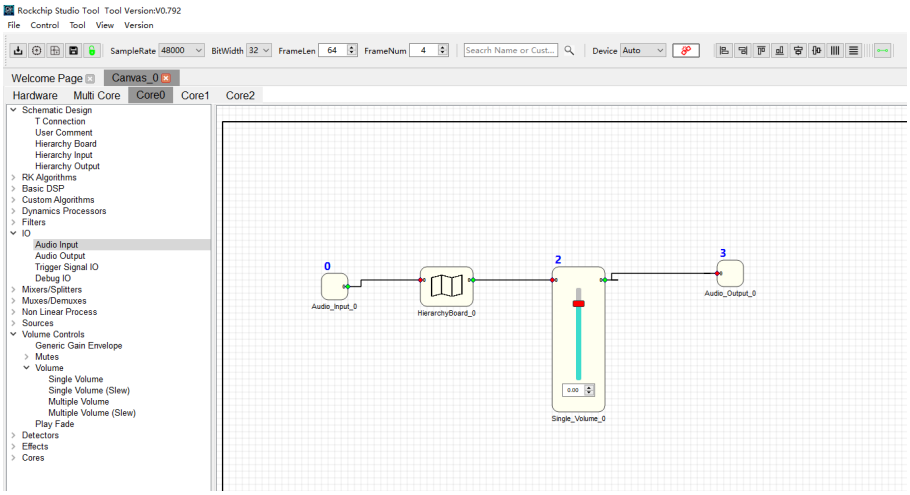


Figure 4.8-1 Algorithm Component Page

4.9 Controls Diagram

The main editing interface, where the selected algorithm components will be displayed. You can drag and layout components left and right, set connections, and double-click to design parameters.

4.10 Information Bar

Records log information, changes to diagrams, and project modifications.

4.11 Element Properties Box

Left-click to select a component, and you can view the basic information of the corresponding component here.

4.12 Sub-diagrams

In the Schematic Design of the component page, the three components related to Hierarchy can be used to create and add input and output interfaces for sub-diagrams. After dragging out the Hierarchy Board, a sub-diagram tab will be added to the sub-diagram bar. At this time, double-clicking the sub-diagram or clicking the corresponding sub-diagram tab will switch to the sub-diagram page. In the main diagram, sub-diagram components are used in the same way as regular components, but new sub-diagrams have no endpoints. You need to drag Hierarchy Input and Hierarchy Output components into the corresponding sub-diagram design page to add input and output endpoints to the sub-diagram components, allowing the sub-diagram to connect with other components.

Note:

- Hierarchy Input: An input port in the main diagram and an output port in the sub-diagram

- Hierarchy Output: An output port in the main diagram and an input port in the sub-diagram

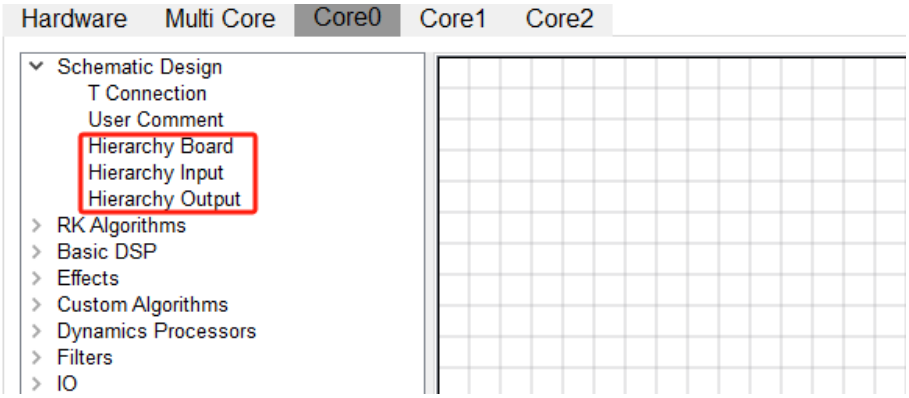


Figure 4.12-1 Sub-diagram Function List

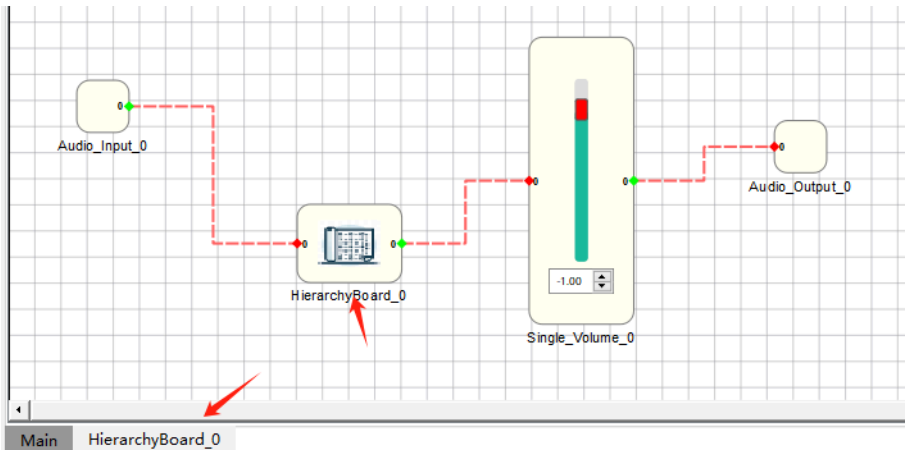


Figure 4.12-2 Creating a New Sub-diagram

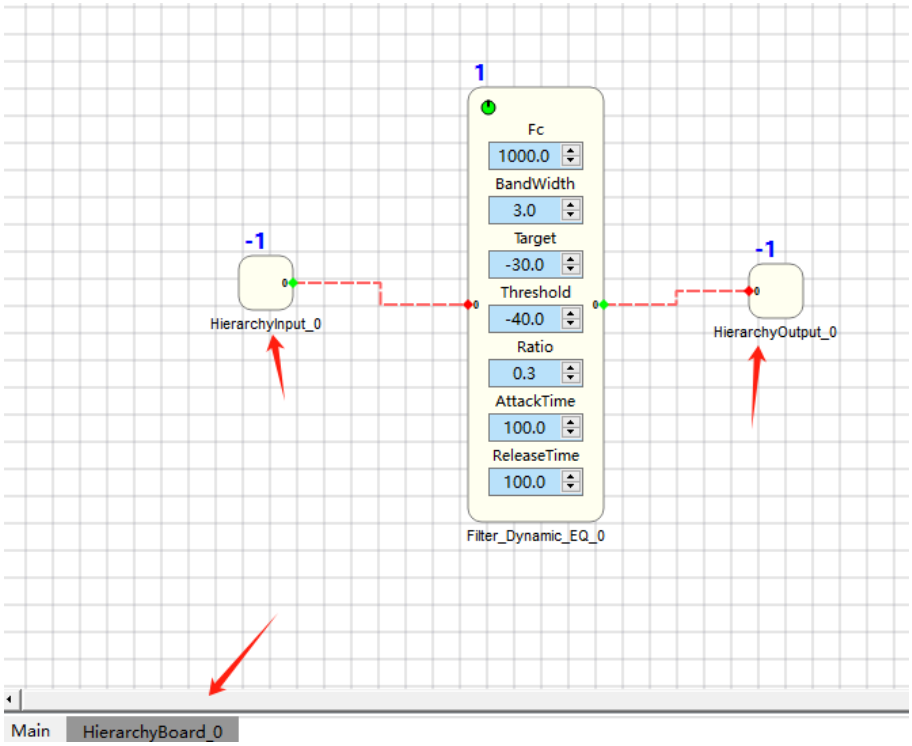


Figure 4.12-3 Adding Inputs and Outputs to a Sub-diagram

4.13 Special Features

4.13.1 Input Port Labeling

Right-click on the component and select Set CHanName

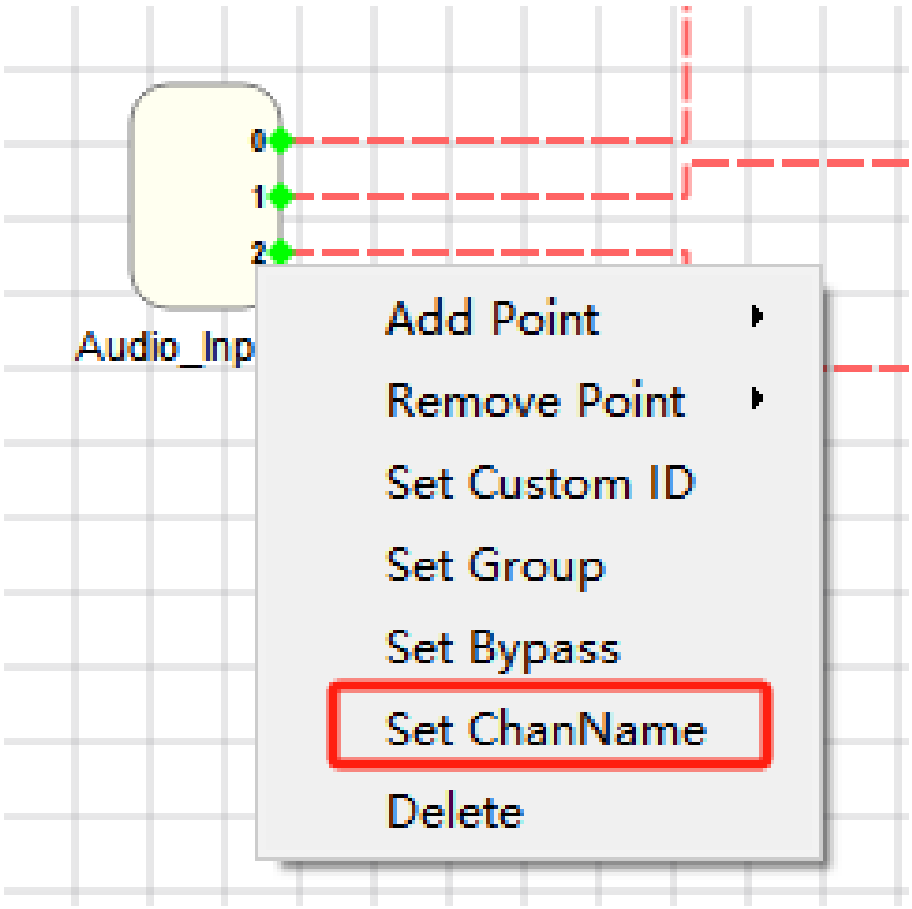


Figure 4.13.1-1 Component

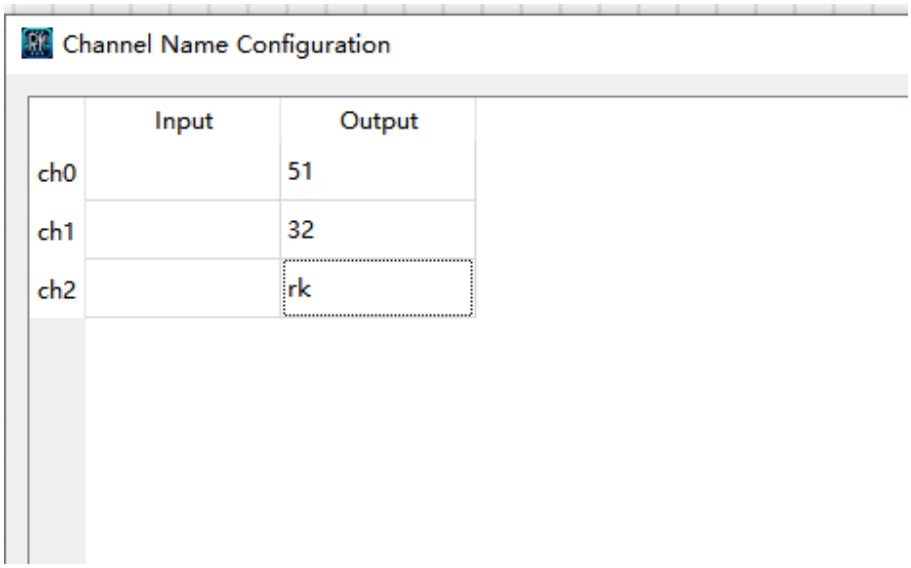


Figure 4.13.1-2 Setting Interface

Here's what it looks like:

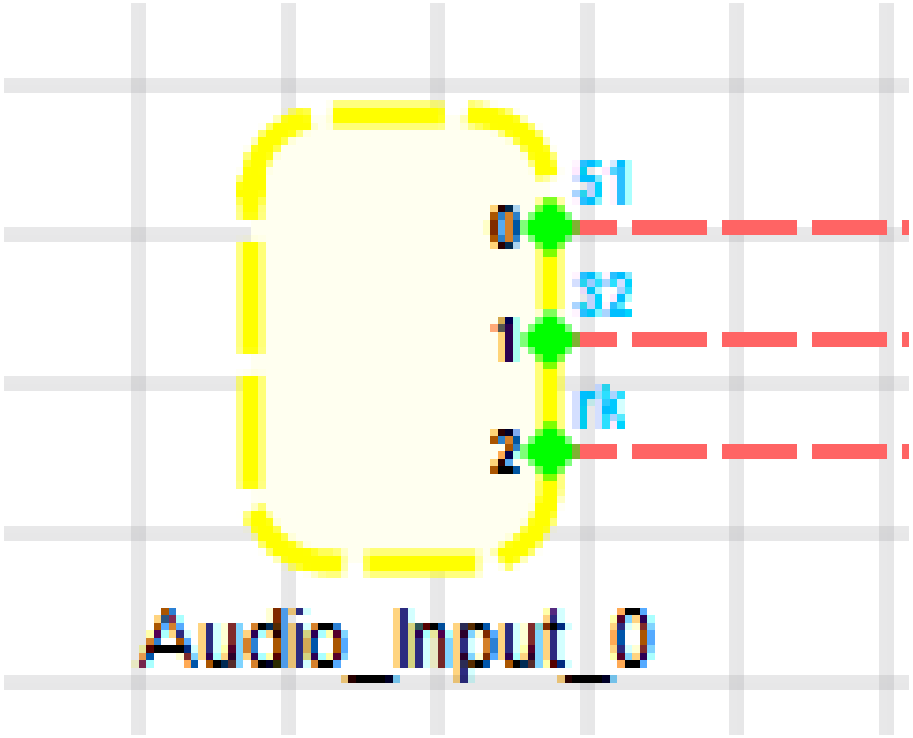


Figure 4.13.1-3 Effect